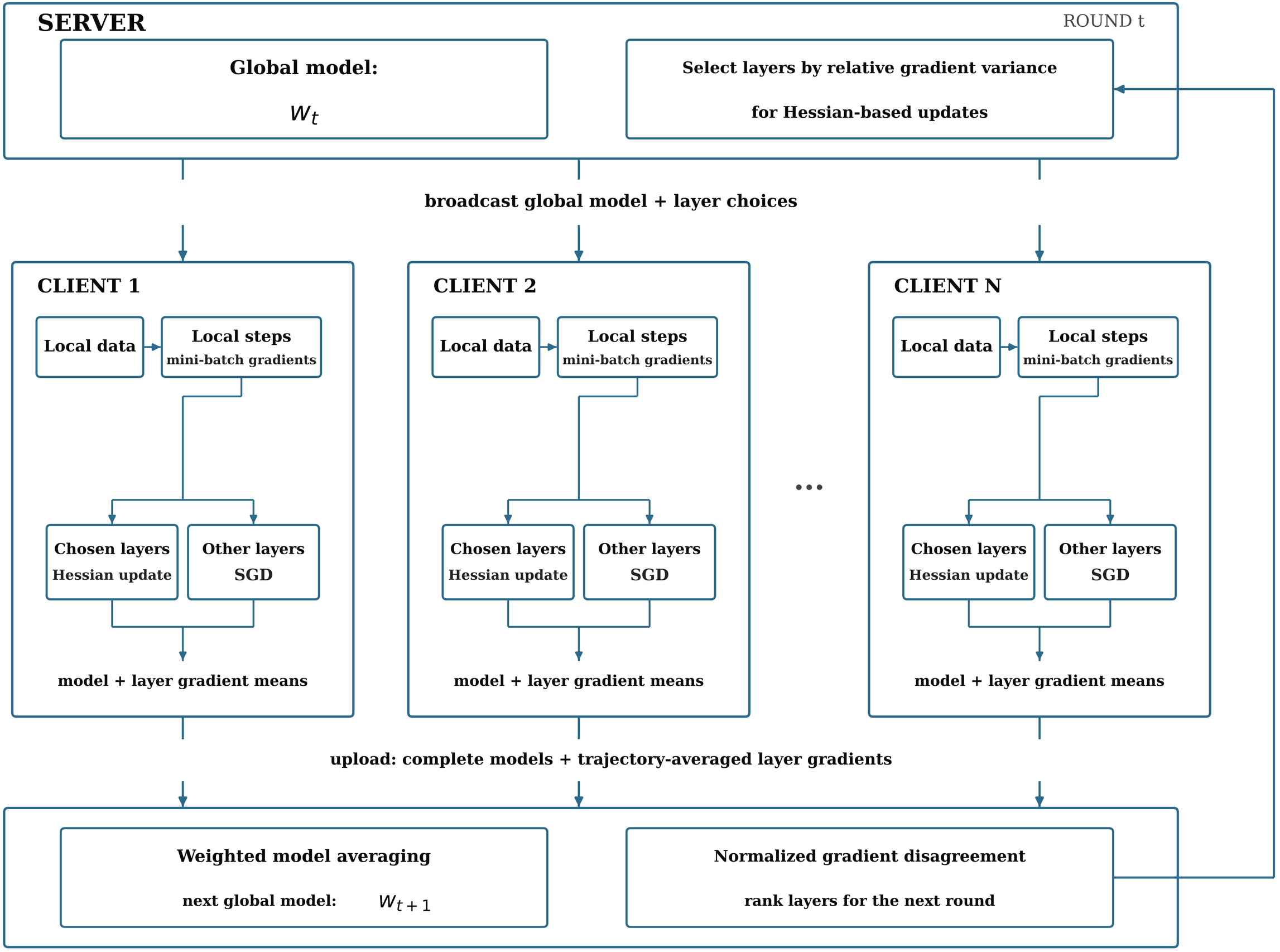
SERVER
ROUND t
Global model:
W_t
Select layers by relative gradient variance
for Hessian-based updates
broadcast global model + layer choices
CLIENT 1
Local data
Local steps
mini-batch gradients
Chosen layers
Hessian update
Other layers
SGD
model + layer gradient means
CLIENT 2
Local data
Local steps
mini-batch gradients
Chosen layers
Hessian update
Other layers
SGD
model + layer gradient means
CLIENT N
Local data
Local steps
mini-batch gradients
Chosen layers
Hessian update
Other layers
SGD
model + layer gradient means
upload: complete models + trajectory-averaged layer gradients
Weighted model averaging
next global model: W_{t+1}
Normalized gradient disagreement
rank layers for the next round